

Pet-Tracker

MSc Electronic Engineering

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Problem Overview

rest in specific spots



knocking over valuable items



jump onto desks with electronic devices or important paper documents



Without human supervision, pets alternate rest and active periods. An untrained pet in restricted areas can cause damage and put itself at risk.

No human supervision



hygiene issues due to shedding



reach exposed food or get too close to appliances



Reference Scenario

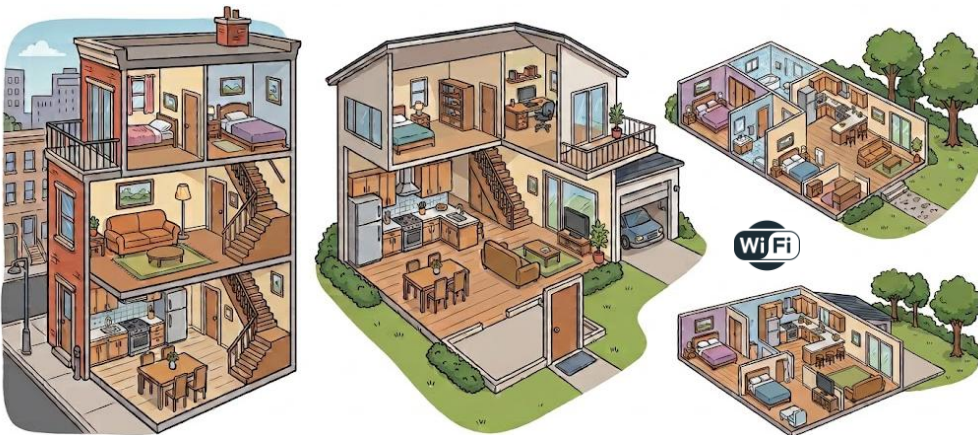
WHO?

Pet owners unable to constantly monitor their pet.



WHERE?

Indoor environments of any size or layout.



WHEN?

During periods of non-supervision.



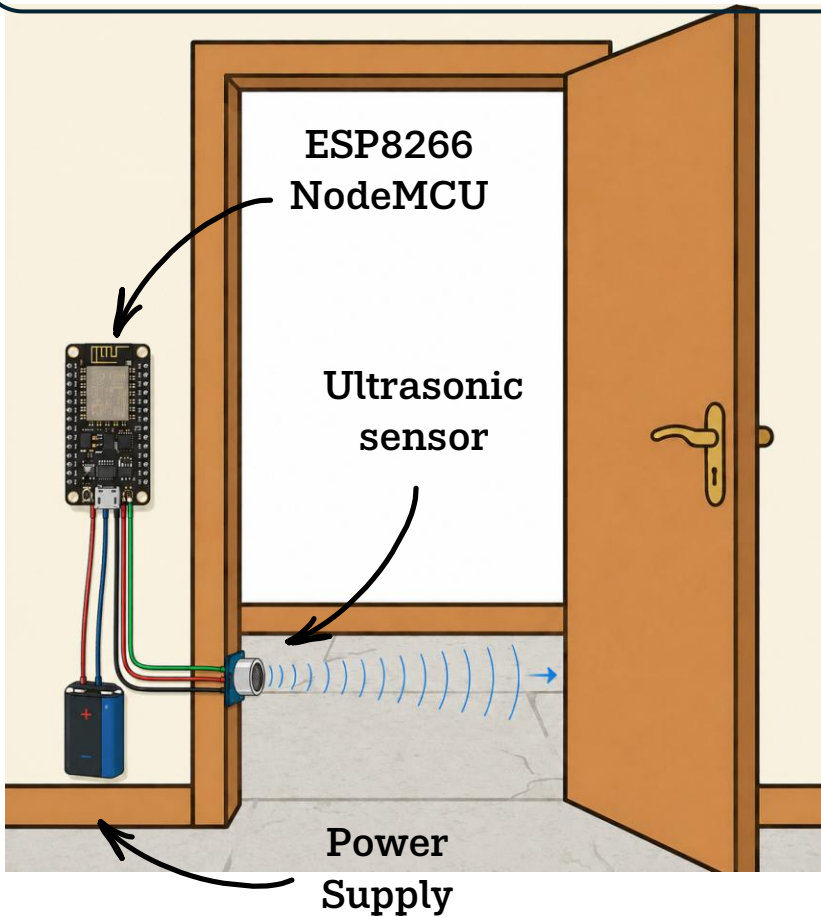
WHAT?

Training purpose & Pet localization



Room transition detection

The door becomes a **smart object**. Each device have to be put in each door to detect **room-to-room transitions**.



After these

HOW can the system distinguish a pet from any other entity?

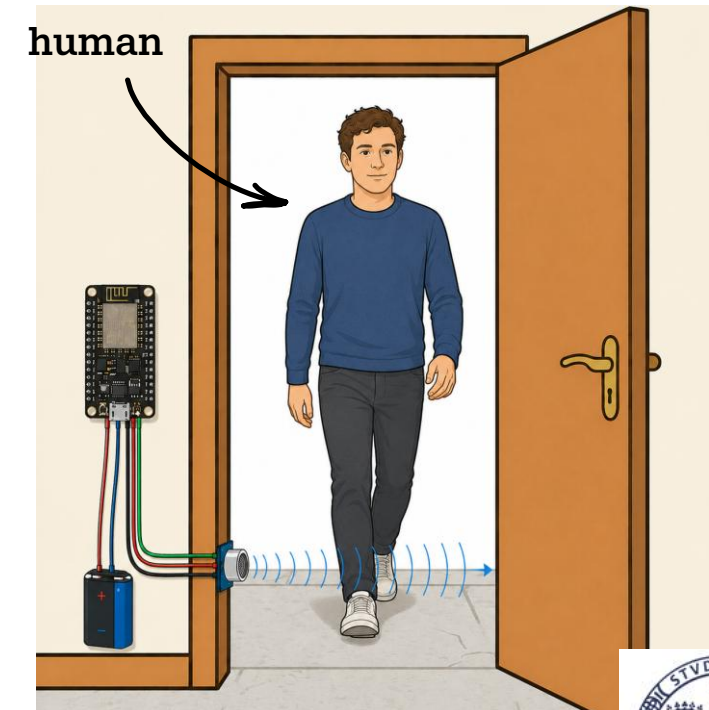
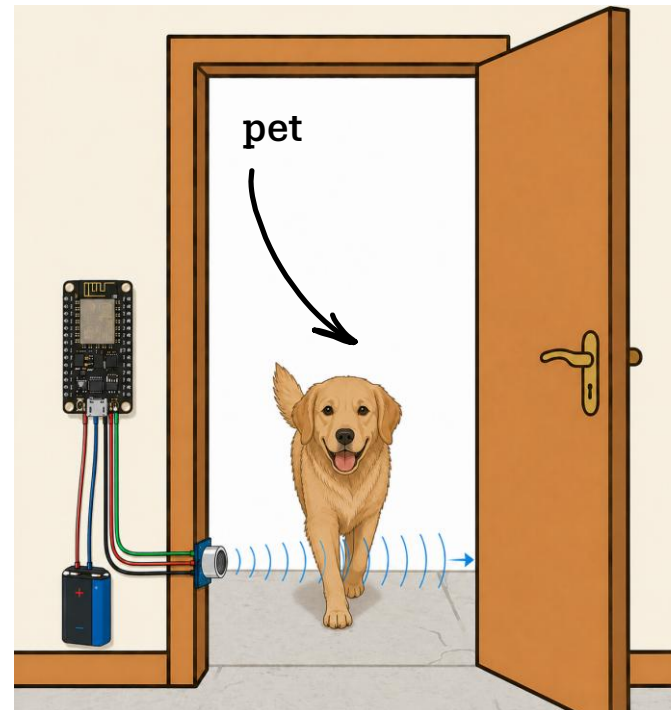
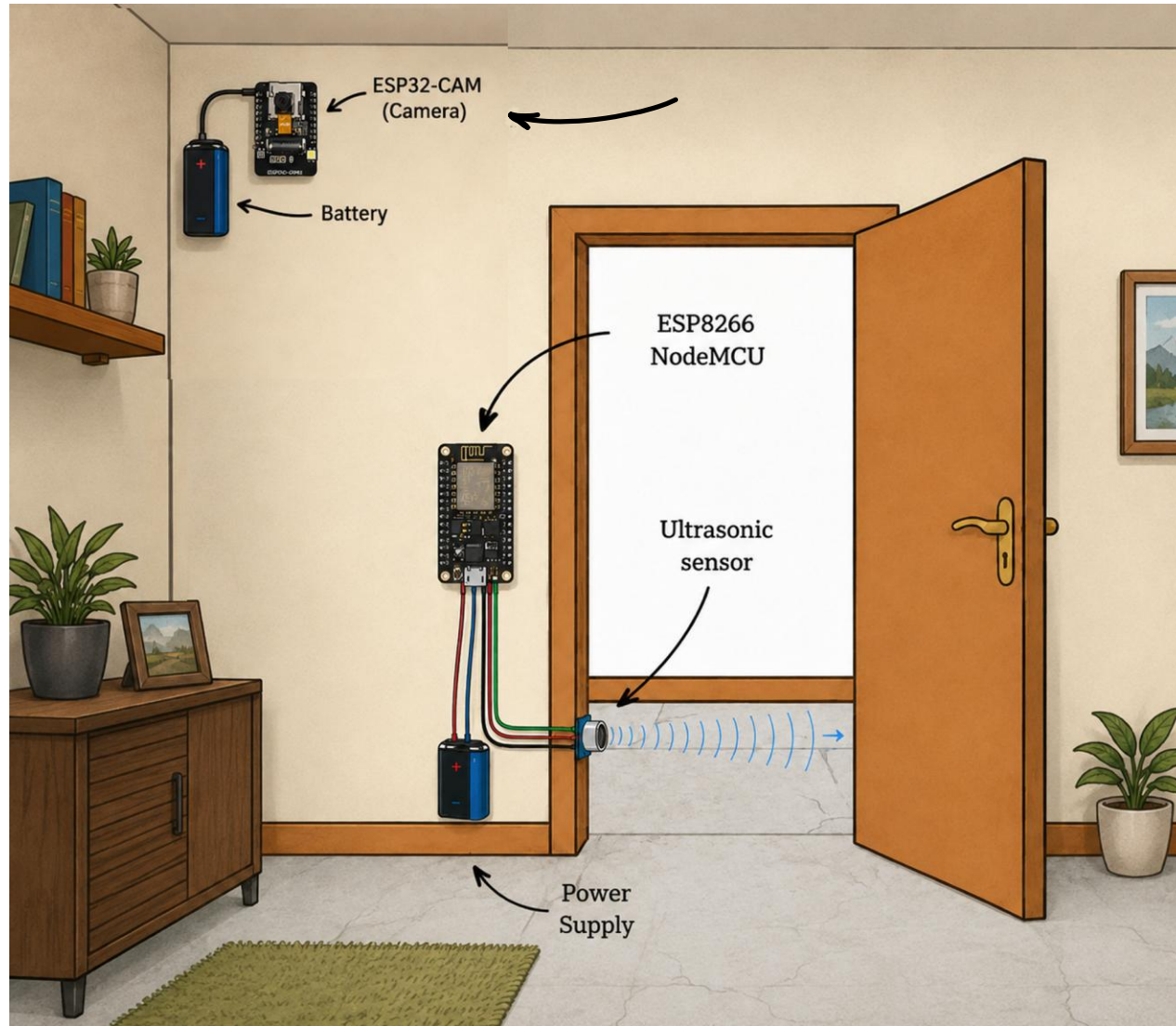
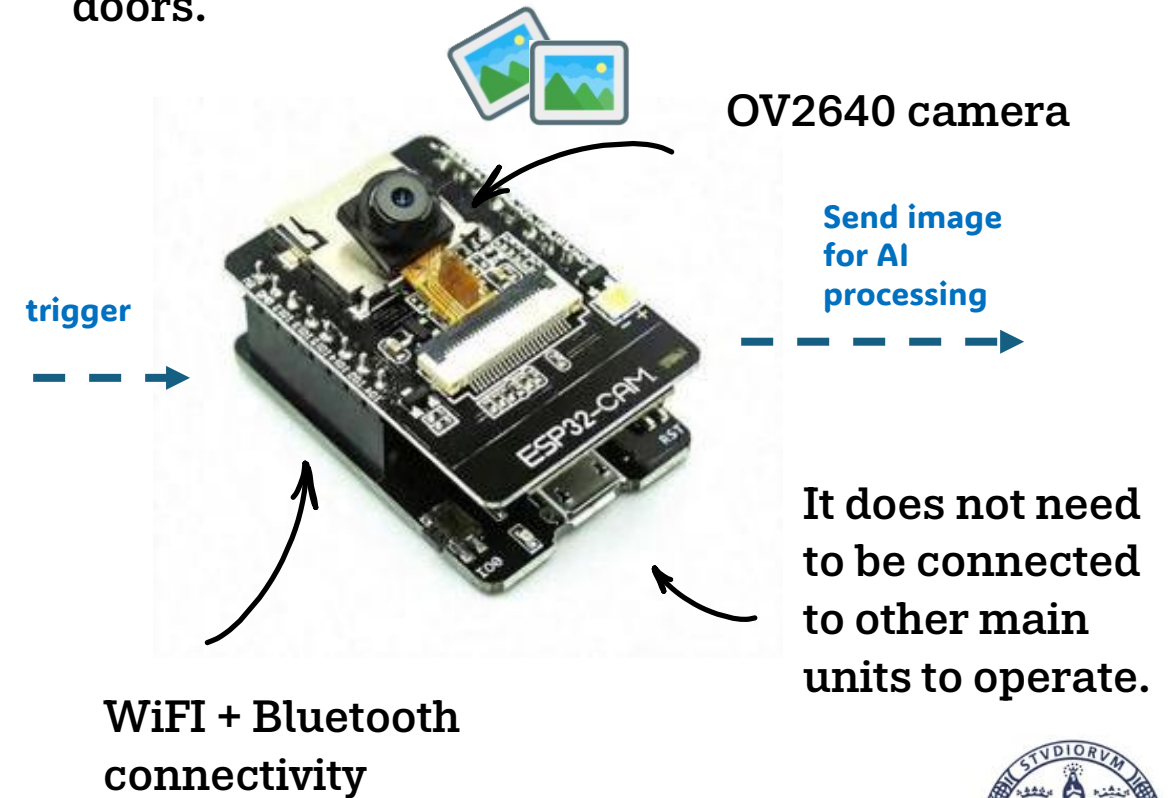


Image acquisition



The room becomes a **smart object**. Each ESP32-CAM has to be placed in each room to **acquire a single image** when triggered by the node upon crossing one of the room's entry doors.



Pet Training System



The pet becomes a **smart object** by attaching the device to either a collar or a harness.

- **Automatic Trigger:** The collar buzzer activates upon the pet entering a restricted room and deactivates upon exit.
- **Manual Override:** The administrator can manually turn the buzzer on or off in any room at any time.



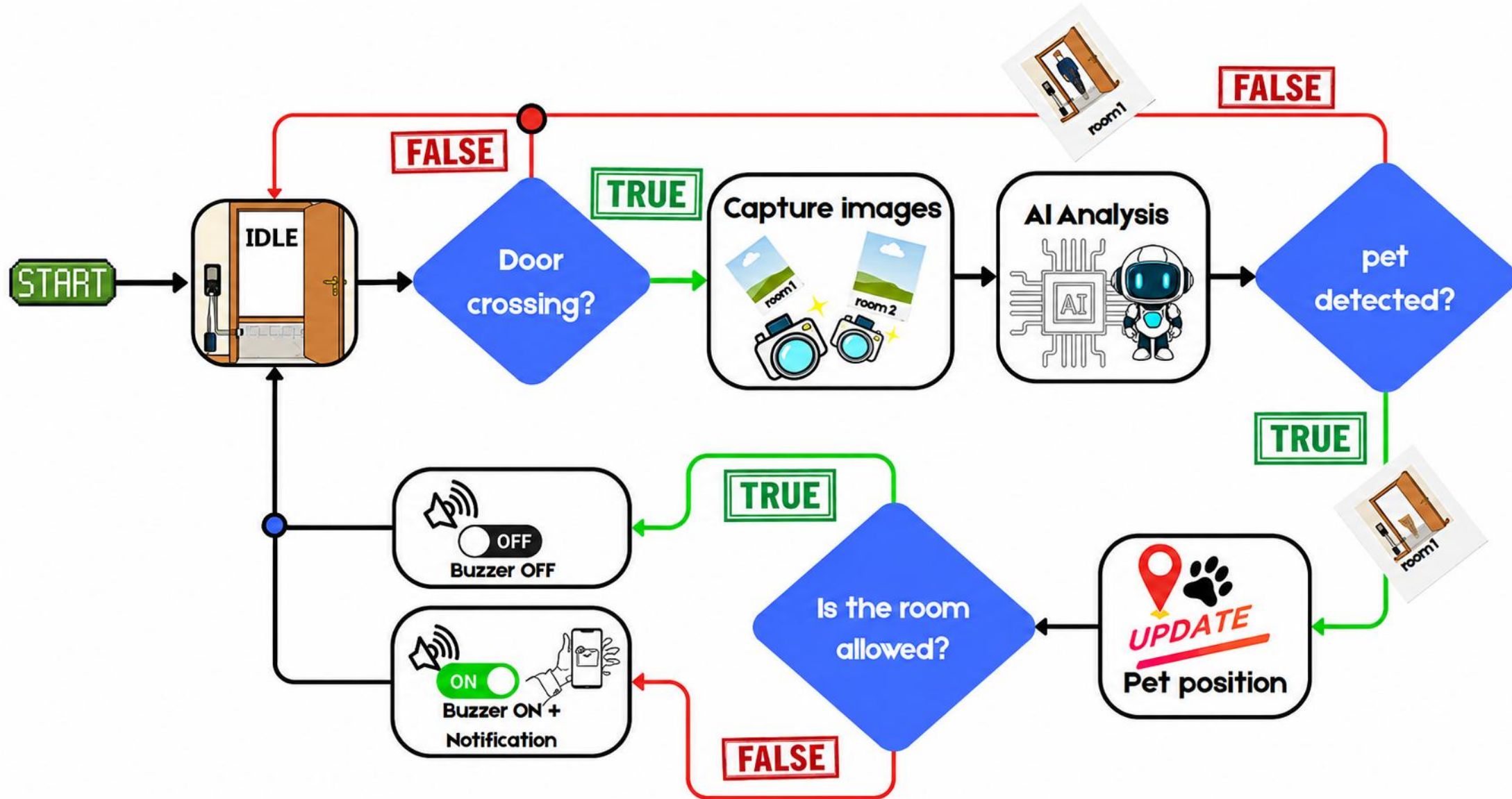
System



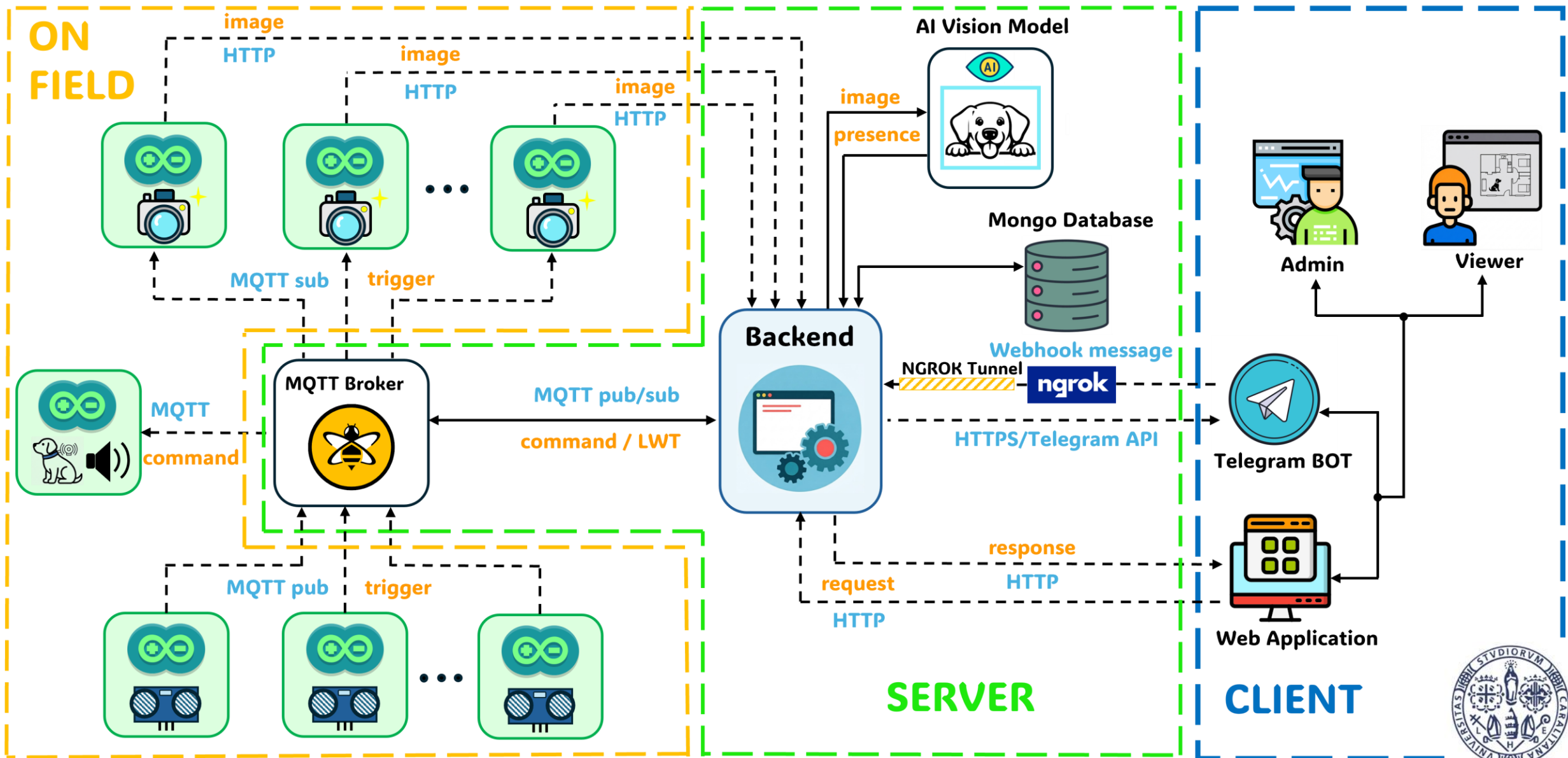
Admin



Workflow of the System



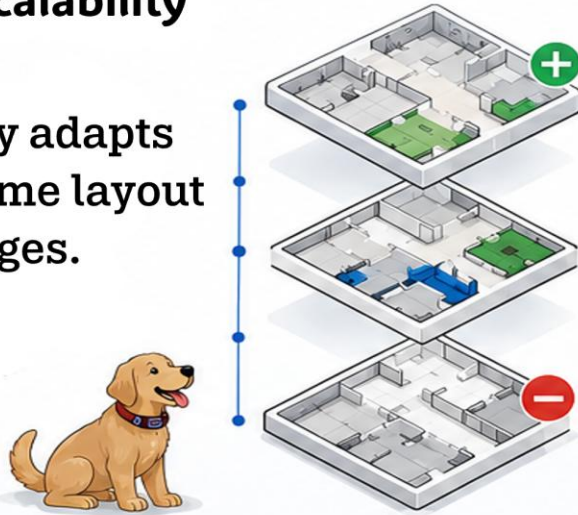
Architecture Overview



Strenght Points of our System

1 Scalability

Easily adapts to home layout changes.



2 Component Replaceability

Easy to replace components in case of failure.



3 System Robustness

Ensures operational continuity even in case of partial failures.



4 Low Latency

Real time user notifications and buzzer activation within 3 s.



5 Privacy

Restricts access to sensitive data.



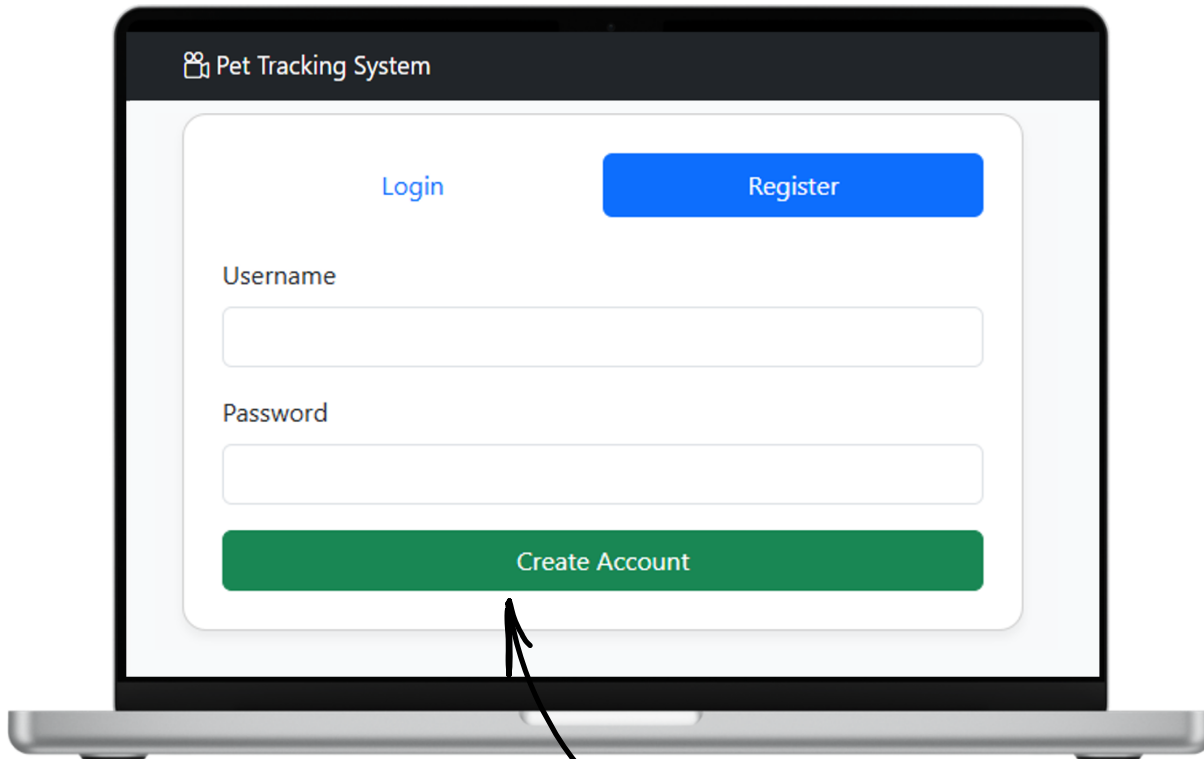
6 User-Friendly Interface

Intuitive UI suitable for everyone.



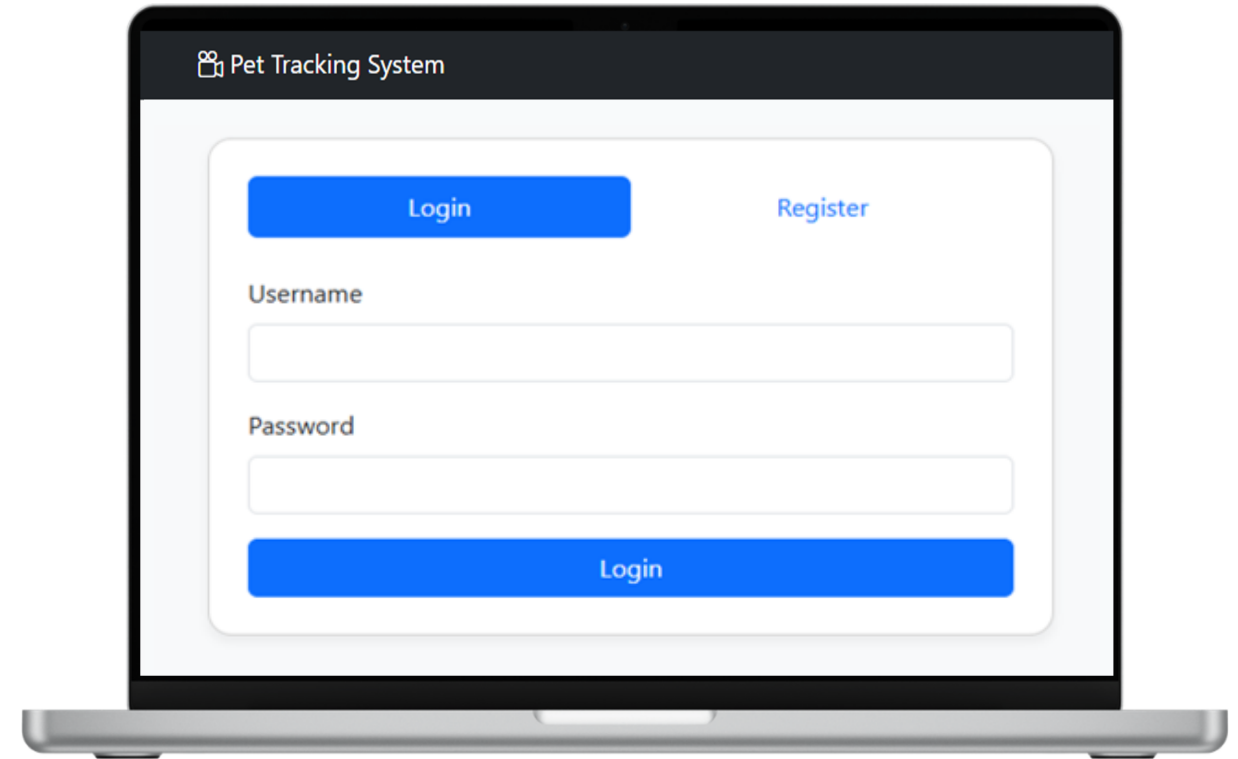
Registration & Login

1° step - **Registration**



Create Account (e.g.
Username: marco138
Password: IoTlexa138!

2° step - **Login**



Your Homes

The screenshot displays the 'Your Homes' section of a 'Pet Tracking System'. The interface includes a header with a user profile icon and a 'Logout' button. Below the header, two virtual environments are listed:

- Sea View Villa** (Admin): Described as 'Relaxing vacation home just steps away from the sandy beach'. It features an 'Enter' button and a 'DELETE' button (red trash icon).
- City Loft** (Viewer): Described as 'Contemporary loft located in the downtown district'. It features an 'Enter →' button.

At the bottom, there is a '+ Create New Home' button.

Annotations:

- Admin Created Home Virtual Environments:** Points to the 'Sea View Villa' entry.
- Viewer Viewable Home Virtual Environments:** Points to the 'City Loft' entry.
- Logout & User profile buttons:** Points to the user profile icon and 'Logout' button in the header.
- DELETE the Home Virtual Environment:** Points to the red trash icon button for 'Sea View Villa'.
- ENTER into the Home Virtual Environment:** Points to the 'Enter' button for 'Sea View Villa' and the 'Enter →' button for 'City Loft'.
- CREATE a new Home Virtual Environment:** Points to the '+ Create New Home' button.



Home Environment Creation

Name
(e.g. Marco's Home)

Description
(e.g. Smart Family Home)

The screenshot shows a laptop screen with the 'Pet Tracking System' header. The main content is titled 'Environment Setup'. Below this, there is a section for 'Step 1: Create Home Environment'. It contains two input fields: 'Name' with the placeholder 'Home Name' and 'Description' with the placeholder 'Brief description'. At the bottom of this section is a blue button labeled 'Create Home'. Two arrows point from the 'Name' and 'Description' labels above to their respective input fields on the screen.

1° step - **Create the Home**

Create Room
(e.g. name: Bathroom,
Floor:1,
Permission: Forbidden)

Create Door
(e.g. name:door_u1)

The screenshot shows a laptop screen with the 'Pet Tracking System' header. The main content is titled 'Step 2: Add Components'. It contains three sections: 'Create Room' with fields for 'Name' (placeholder: e.g. Kitchen), 'Floor' (placeholder: 0), and 'Permission' (dropdown menu showing 'Forbidden' and a green '+' button); 'Create Door' with a 'Name' field (placeholder: e.g. Front Door) and a green '+' button; and 'Create Pet' with a 'Name' field (placeholder: e.g. Fido) and a yellow '+' button. Below these sections is a summary table titled 'Components added so far:' with columns for 'Rooms', 'Doors', and 'Pets', each showing 'None yet'. At the bottom is a blue button labeled 'Go to Dashboard'. Three arrows point from the 'Create Room', 'Create Door', and 'Create Pet' labels above to their respective sections on the screen.

Create Pet
(e.g. name: Fido)

2° step - **Add Components**



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Dashboard

After clicking the **Enter →** button in *Your Homes*

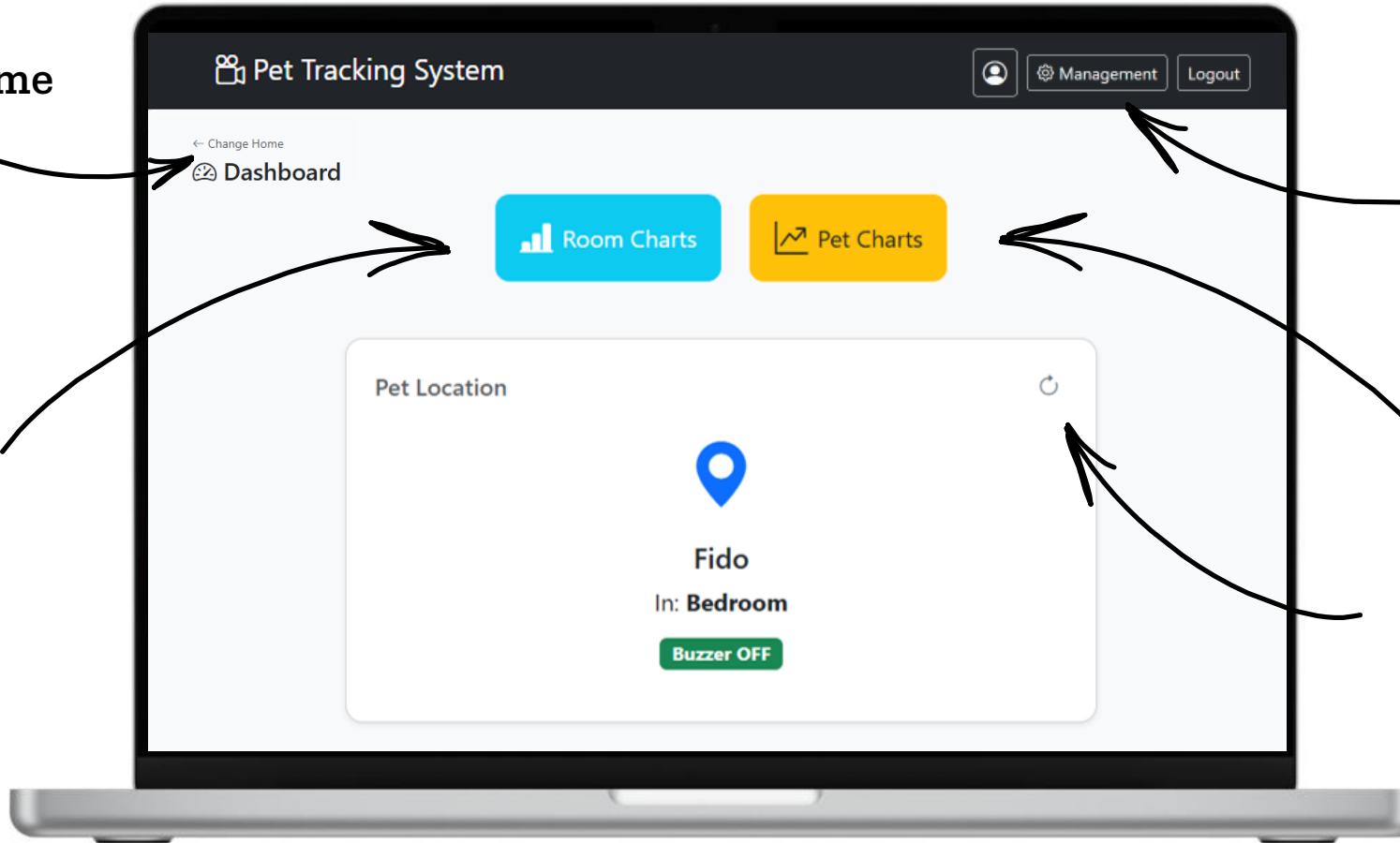
Button visible only to the administrator (used to update rooms, doors, pet configuration and to add/remove viewers)

Change Home

Daily Room Statistics

Statistics on pet violations for the most recent 30 days

Query the system for the pet current position



Management

The screenshot shows the 'Pet Tracking System' Management interface. At the top, there's a header with the system name, a user profile icon, and buttons for 'Management' and 'Logout'. Below the header, a 'Home Management' section contains a 'Home ID' field with a 'Show ID' button. The main area is divided into four panels: 'Rooms', 'Doors', 'Pet', and 'Viewers'. The 'Rooms' panel lists rooms with their current state (allowed/forbidden) and a 'Show ID' button. The 'Doors' panel shows a list of doors with a 'Show ID' button and a green '+' button. The 'Pet' panel shows a list of pets with a 'Show ID' button and a green '+' button. The 'Viewers' panel shows a list of viewers with a 'Show ID' button and a green '+' button. Annotations with arrows point to specific elements: 'Show Room ID button' points to the 'Show ID' button in the Rooms panel; 'Change the state of the room permission' points to the dropdown menu in the Rooms panel; 'Update pet configuration' points to the 'Show ID' button in the Pet panel; 'Update door configurations' points to the green '+' button in the Doors panel; and 'Add /Remove Viewers' points to the green '+' button in the Viewers panel.

Pet Tracking System

Management Logout

← Back to Dashboard

Home Management

Home ID Show ID

Rooms

Room	State	Permission	Action
Kitchen	allowed	Allowed	Show ID Allowed ✓ ✕
Living room	allowed	Allowed	Show ID Allowed ✓ ✕
Bedroom	forbidden	Forbidden	Show ID Forbidden ✓ ✕
Office	allowed	Allowed	Show ID Allowed ✓ ✕
Bathroom	allowed	Allowed	Show ID Allowed ✓ ✕

Name

Floor Permission Allowed ✓ ✕

Doors

✕

Name +

Pet

✕

Name +

Viewers

✕

Username +

Show Room
ID button

Change the
state of the
room
permission

Update door
configurations

Add /Remove
Viewers

Update pet
configuration



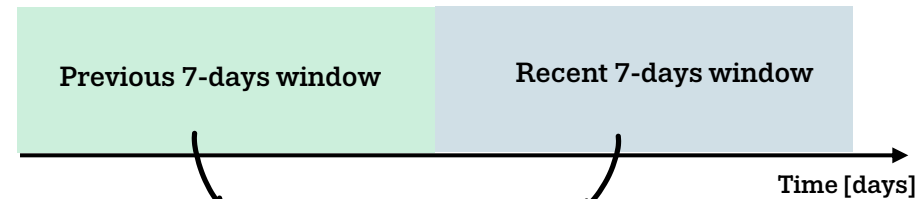
Pet Charts

Forbidden Room Entries per day

Average buzzer time [min] per day

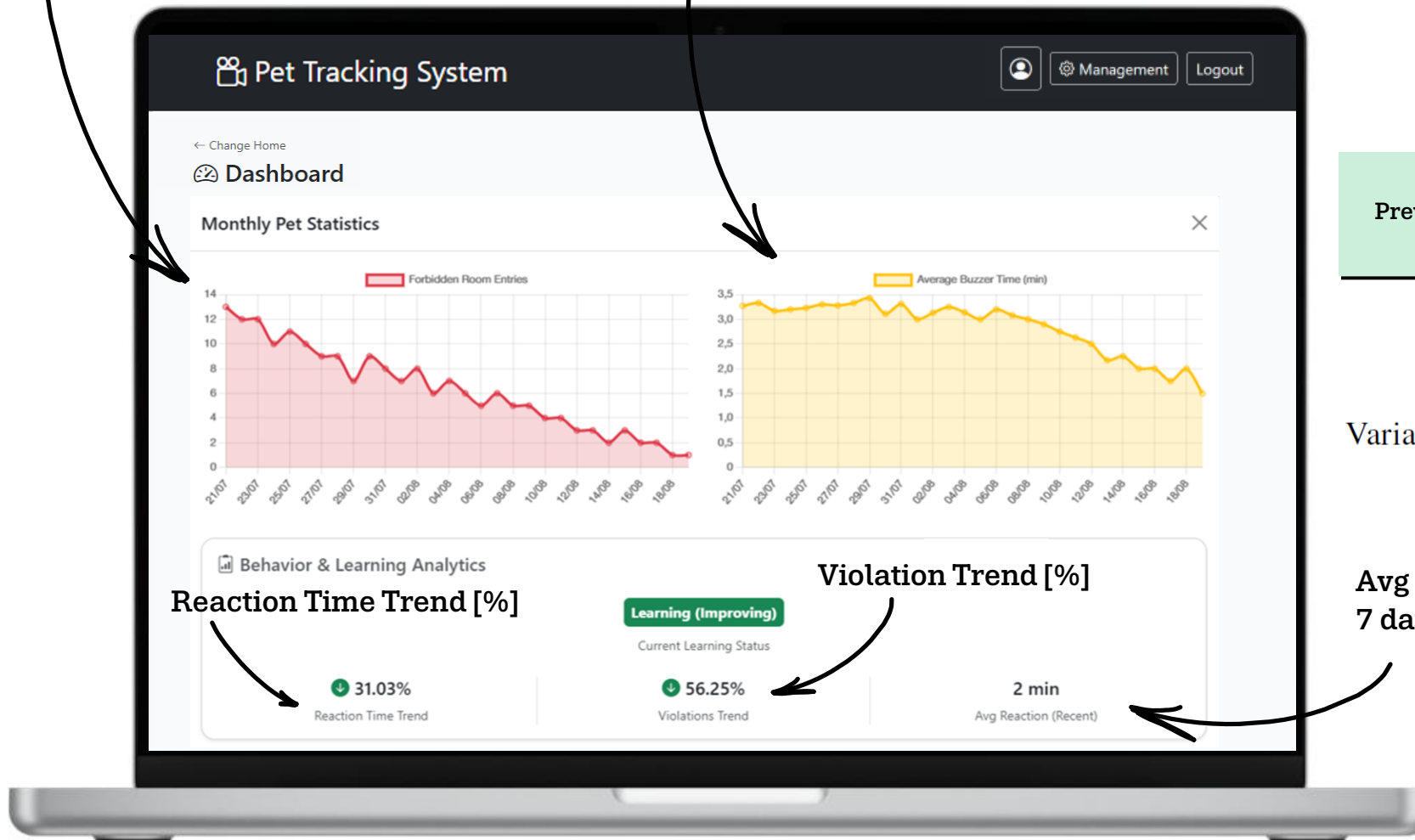
$$\text{Daily Average Reaction Time} = \frac{\text{Daily Buzzer Duration (mins)}}{\text{Daily Violations Count}}$$

DATA ANALISYS



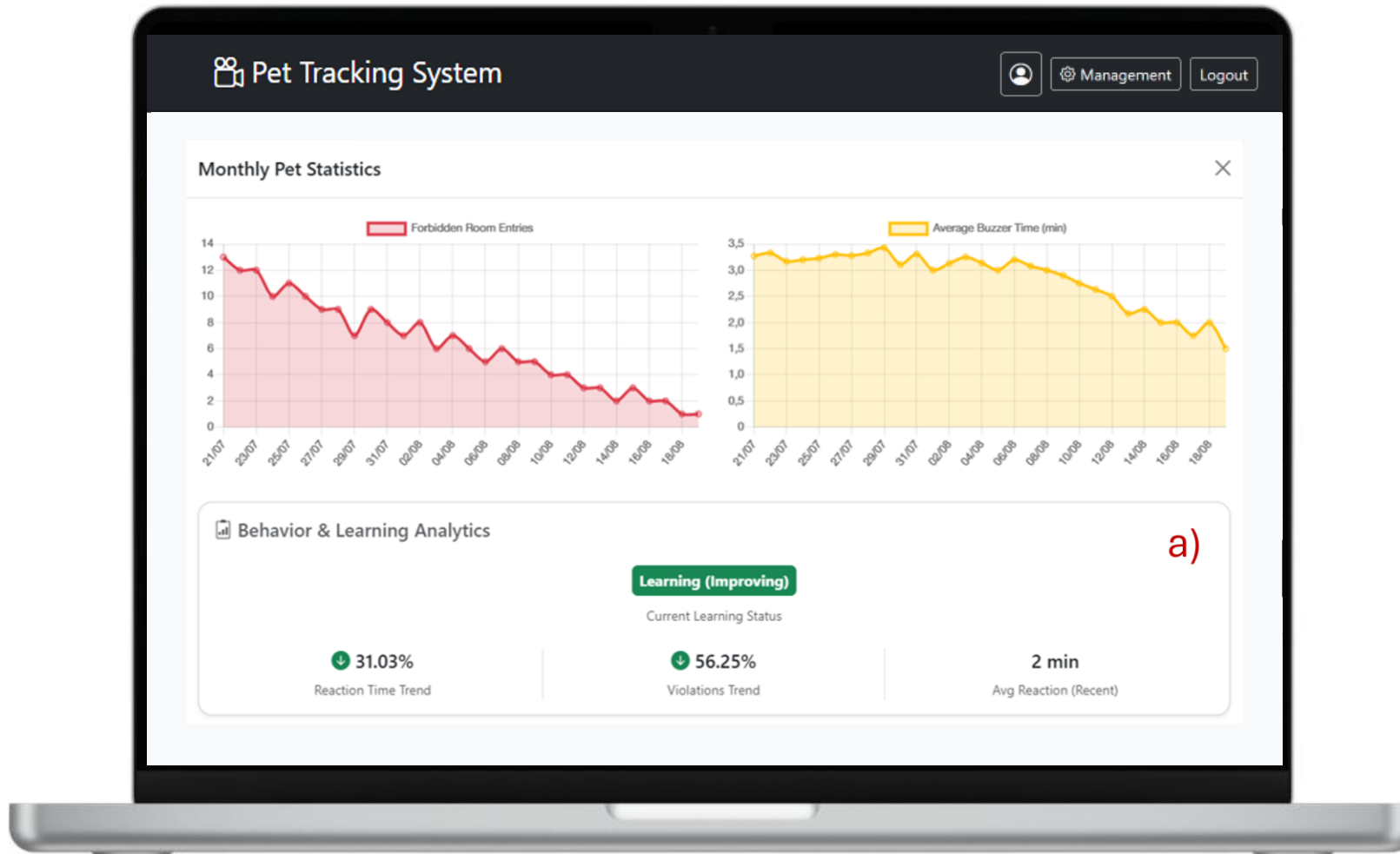
$$\text{Variation \%} = \left(\frac{\text{Recent Value} - \text{Previous Value}}{\text{Previous Value}} \right) \times 100$$

Avg Reaction (Recent 7 days windows) [min]



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Learning (Improving)

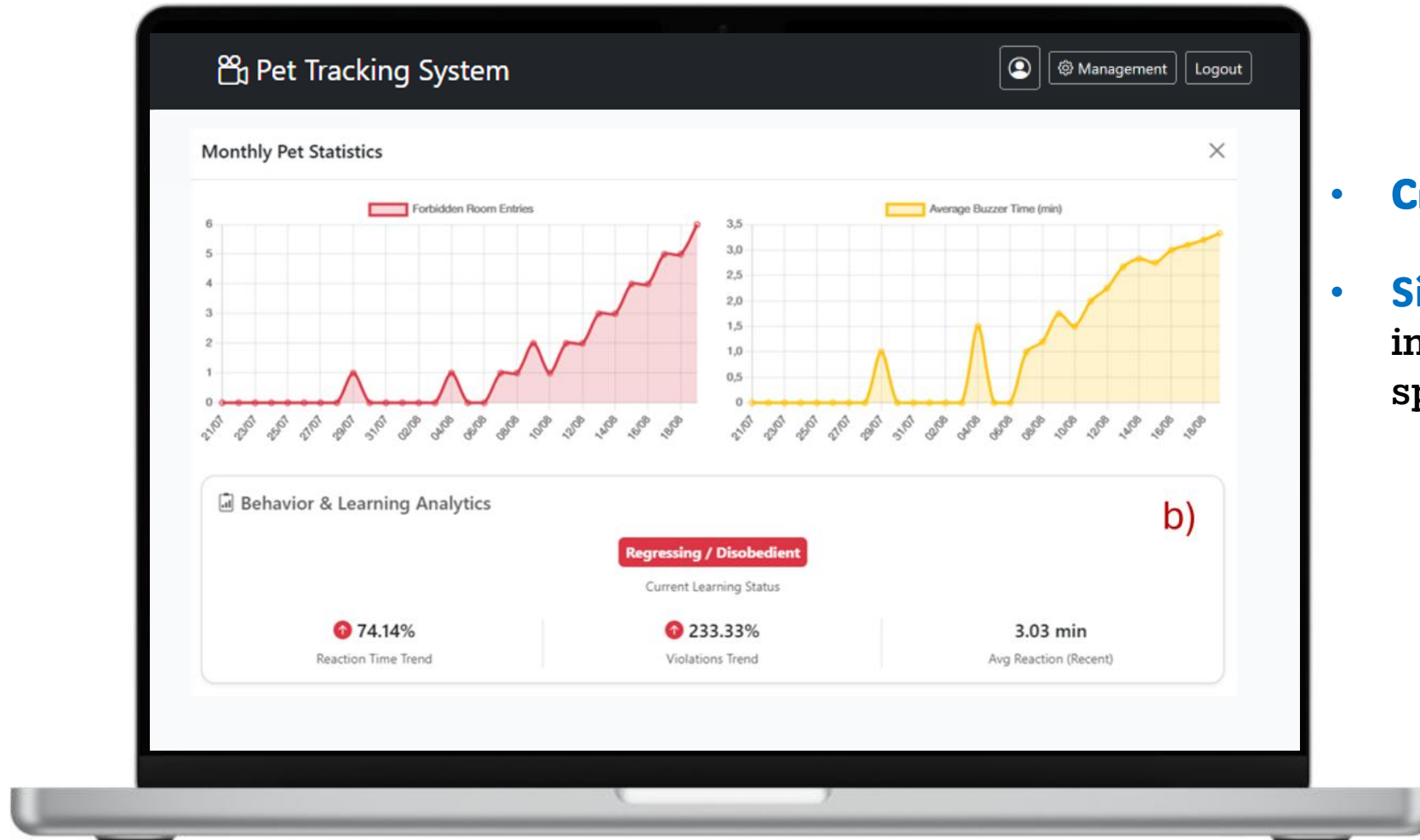


Criterion: Variation $< -10\%$

Significance: Positive response to deterrent; clear understanding of boundaries.



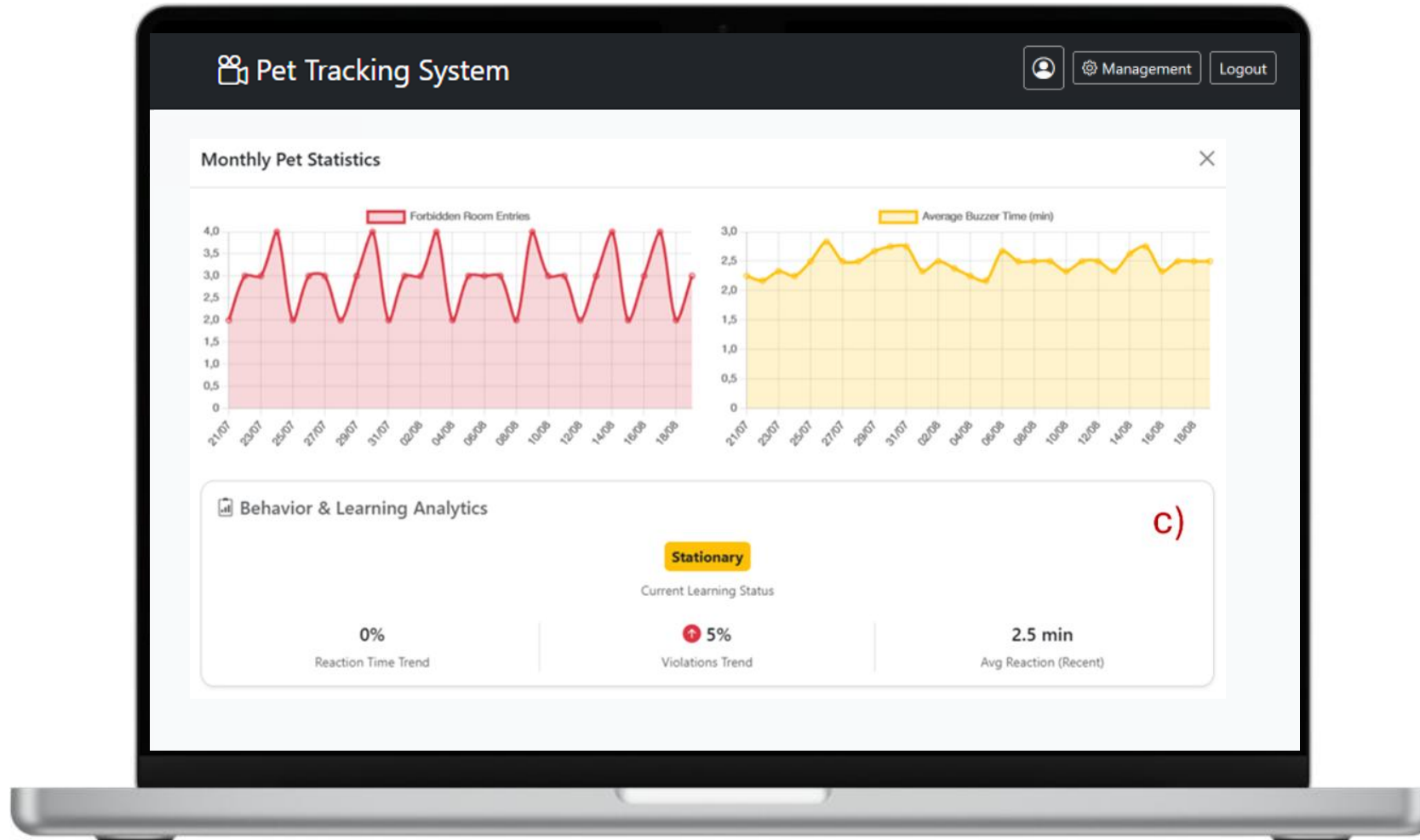
Regressing/Disobedient



- **Criterion:** Variation $> 10\%$
- **Significance:** Performance decline; increased tendency to enter forbidden spaces.



Stationary

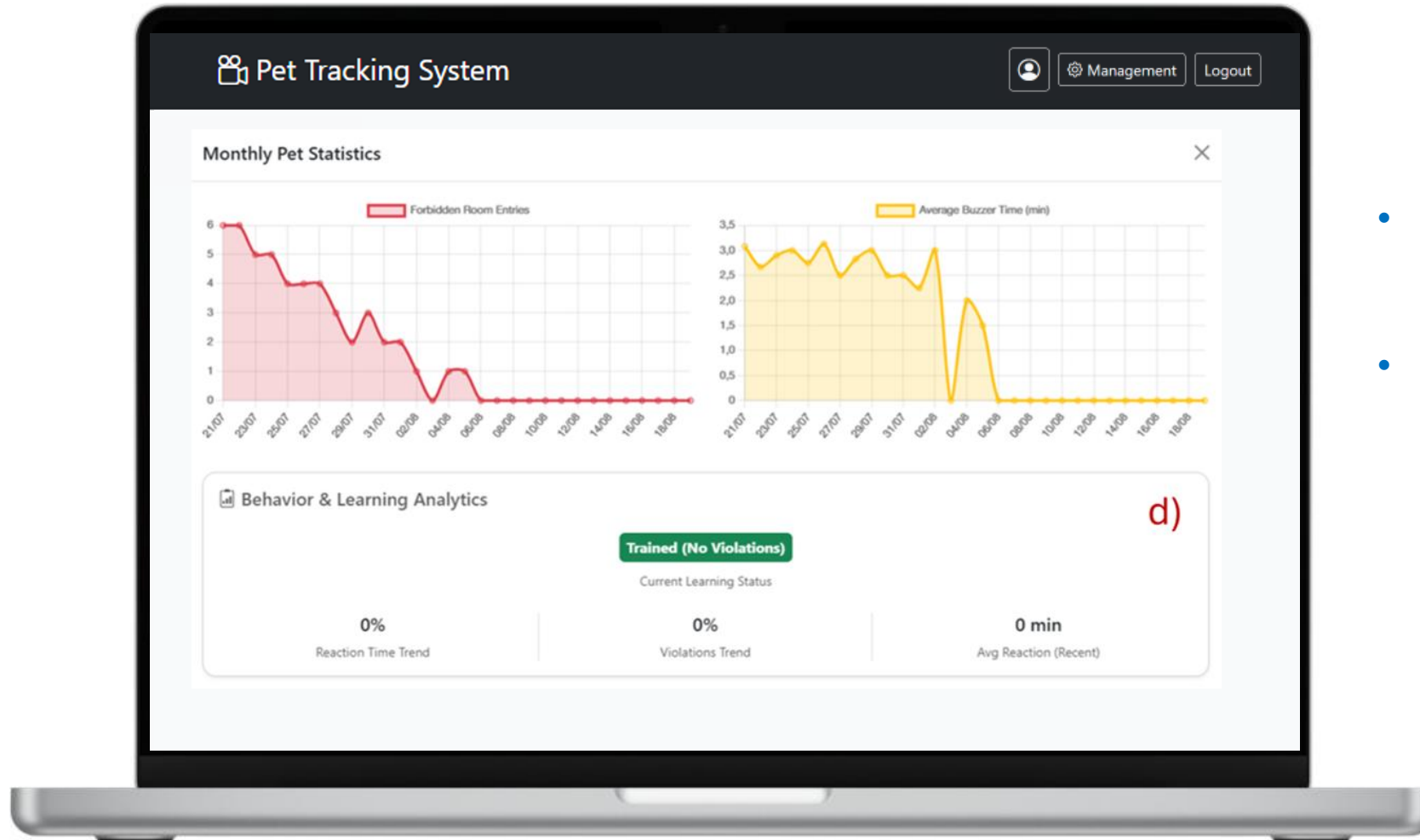


Criterion: Variation within $[-10\%, +10\%]$ threshold.

Significance: Stable performance; no significant improvement or deterioration.



Trained (no Violations)

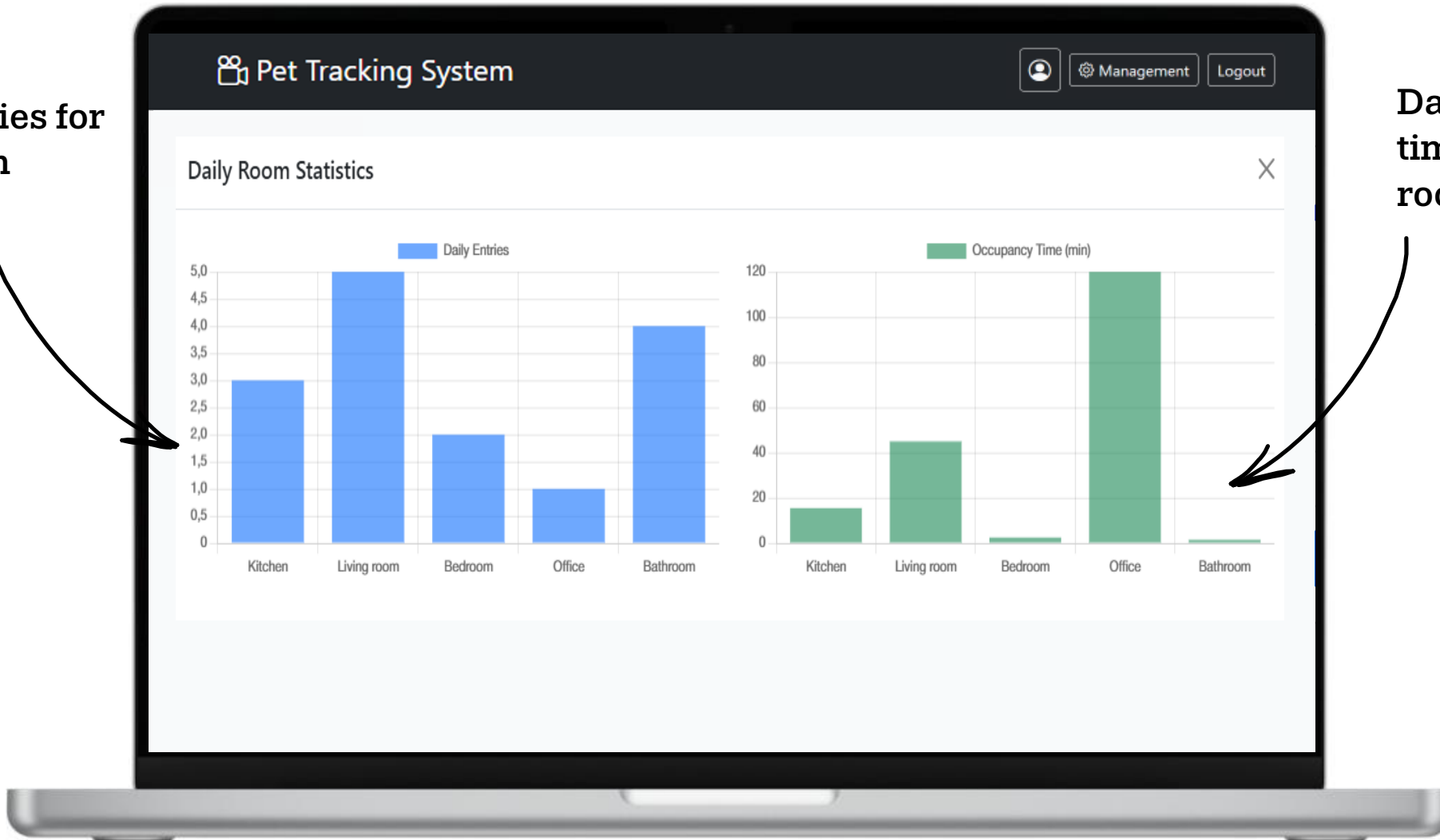


- **Criterion:** Zero violations in the recent window.
- **Significance:** Successful training; consistent respect for restricted boundaries.



Room Charts

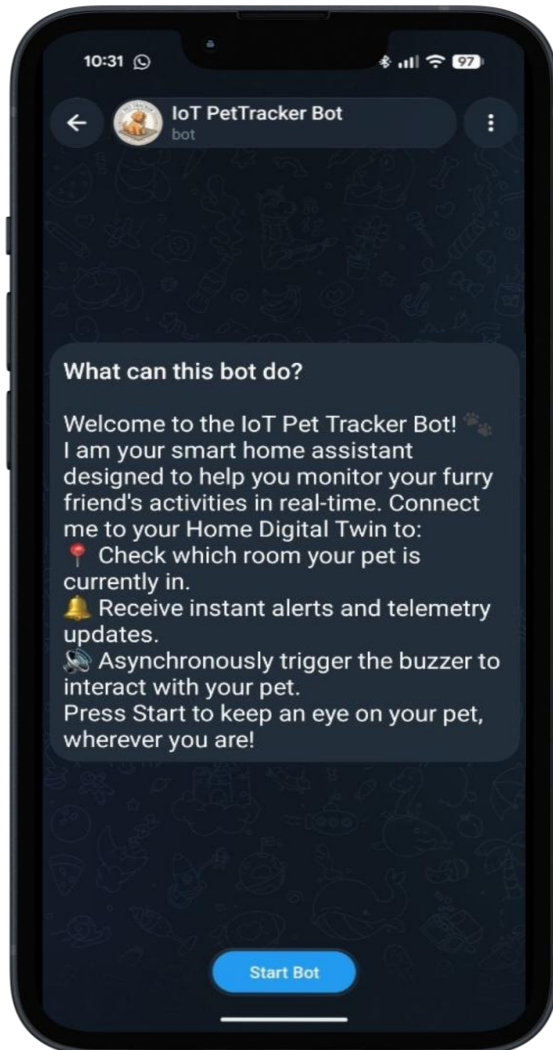
Daily entries for each room



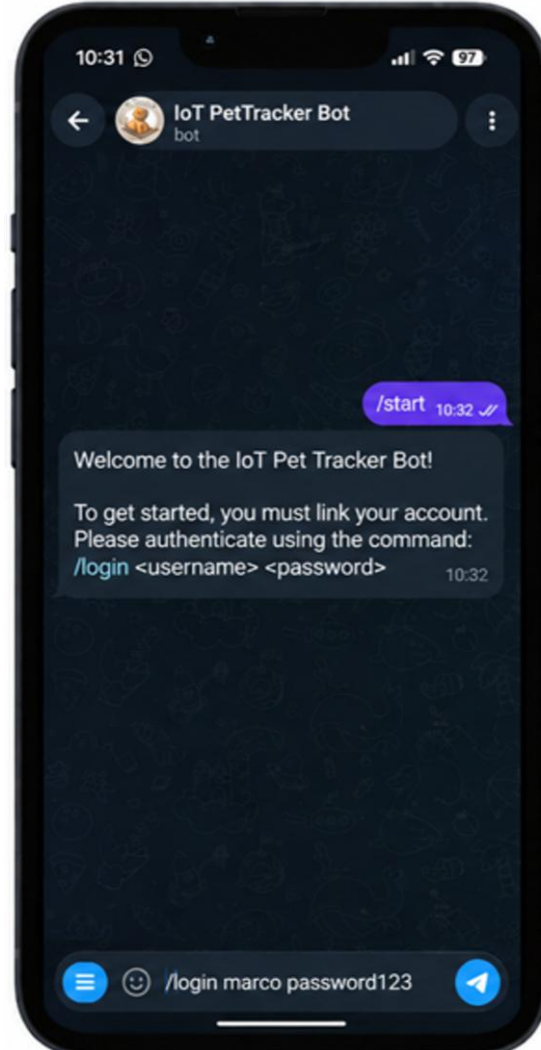
Daily occupancy time for each room [min]

The Telegram Bot- Start Bot and login

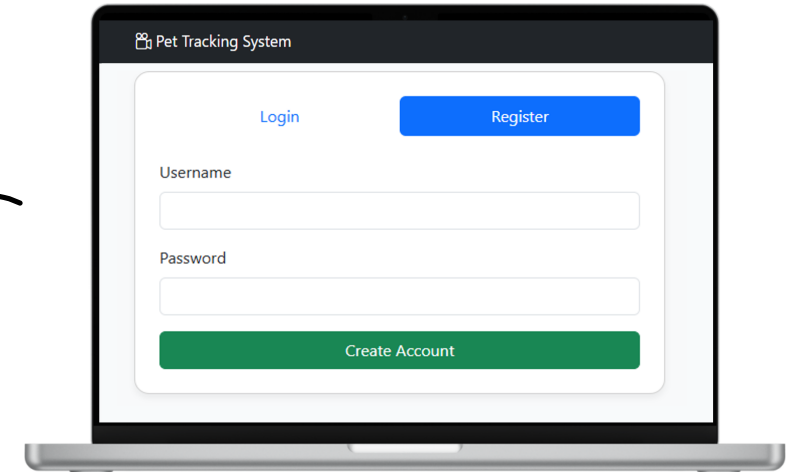
Main informations



2° step - /login



1° step - Registration



next

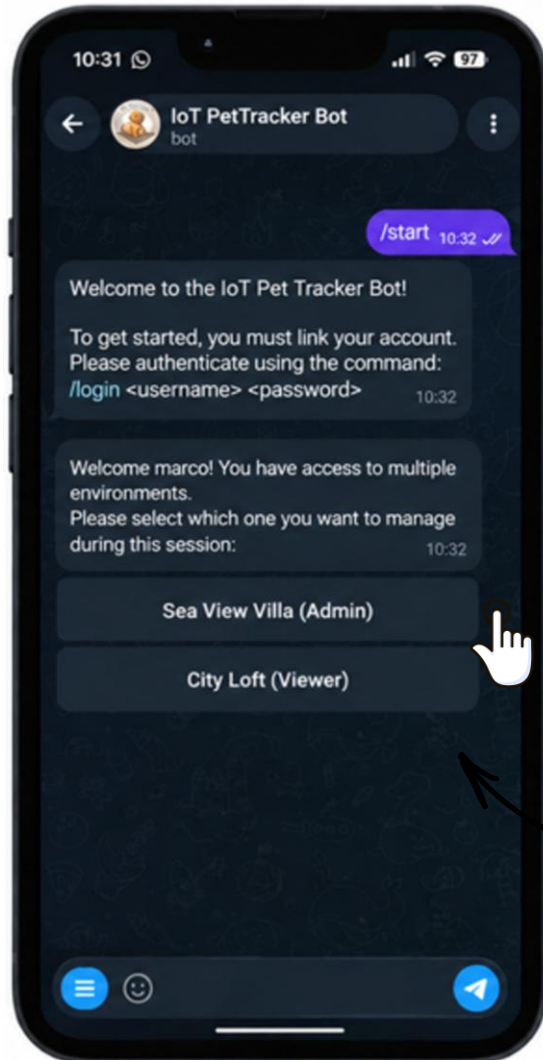
Account registration on the Web App is required before logging into the Telegram Bot (use the same credentials).



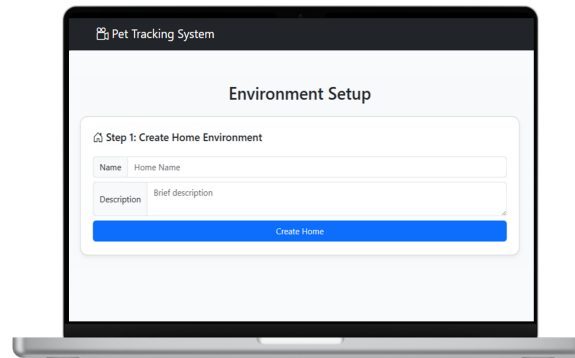
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The Telegram Bot- Home Selection and logout

2° step - Home selection

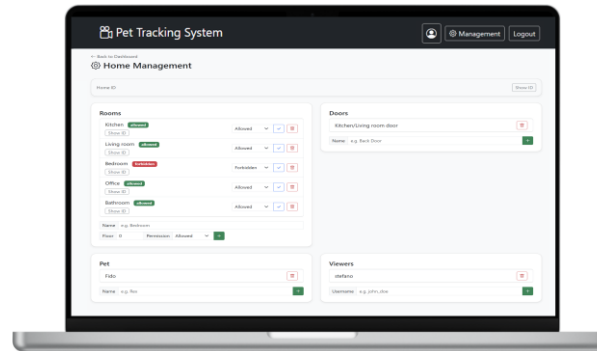


1° step - Create an Home



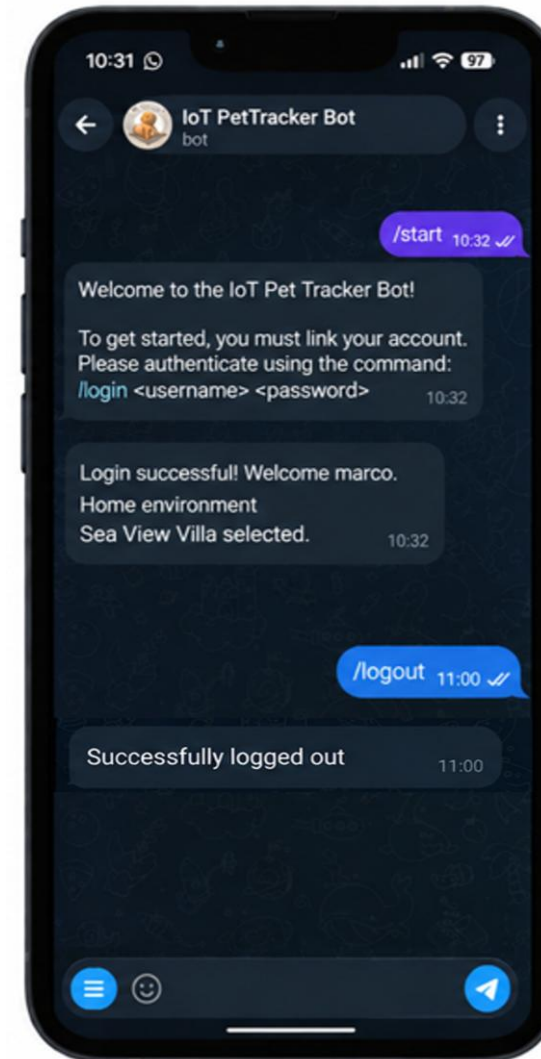
OR

Be added as Viewer



Select an accessible environment based on user role (Admin or Viewer).

/logout



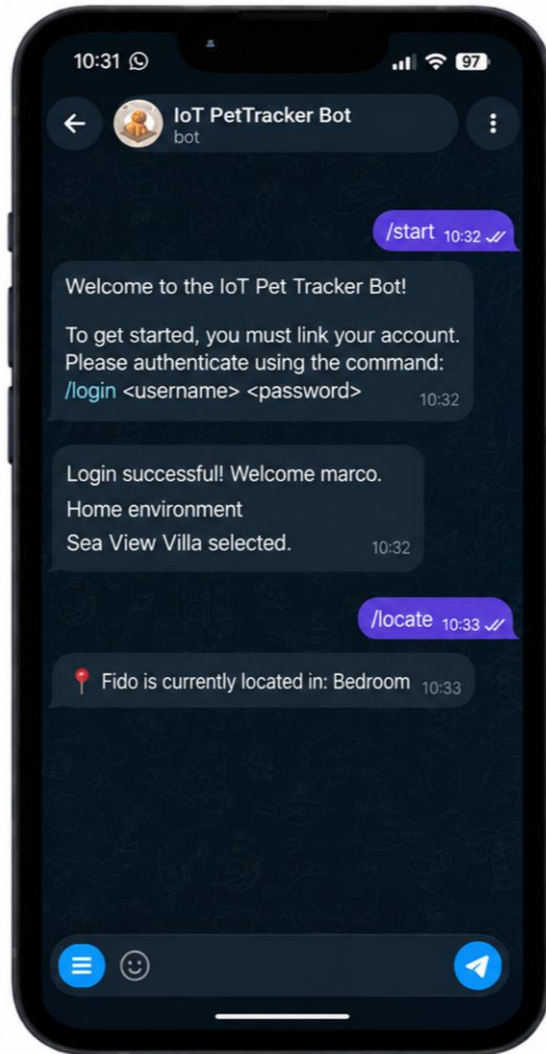
Users can explicitly end their session at any time



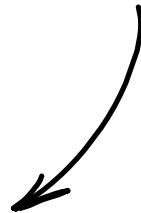
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The Telegram Bot- \locate and \buzzer commands

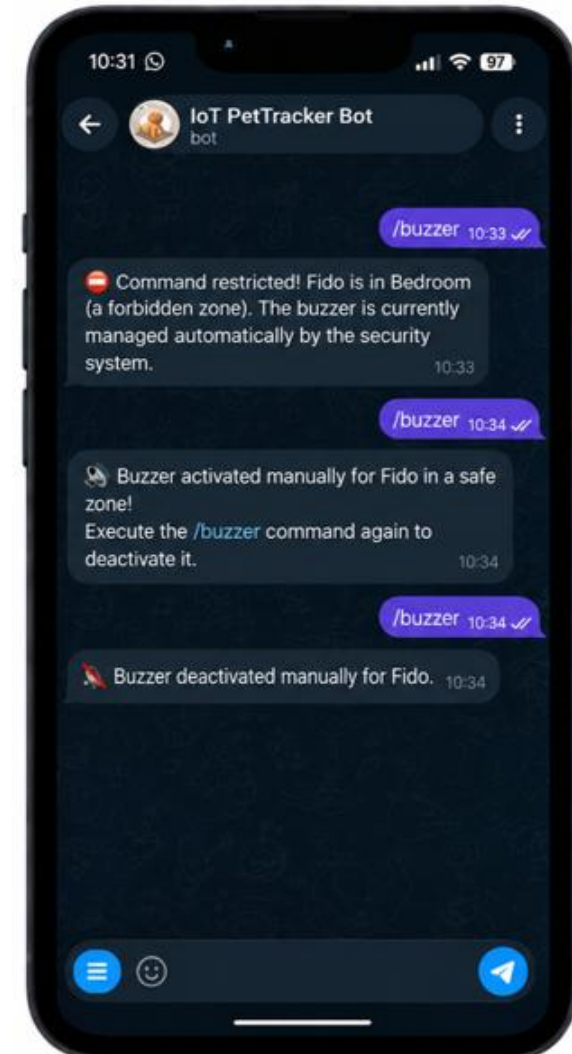
/locate



Query the system
for real-time pet
position



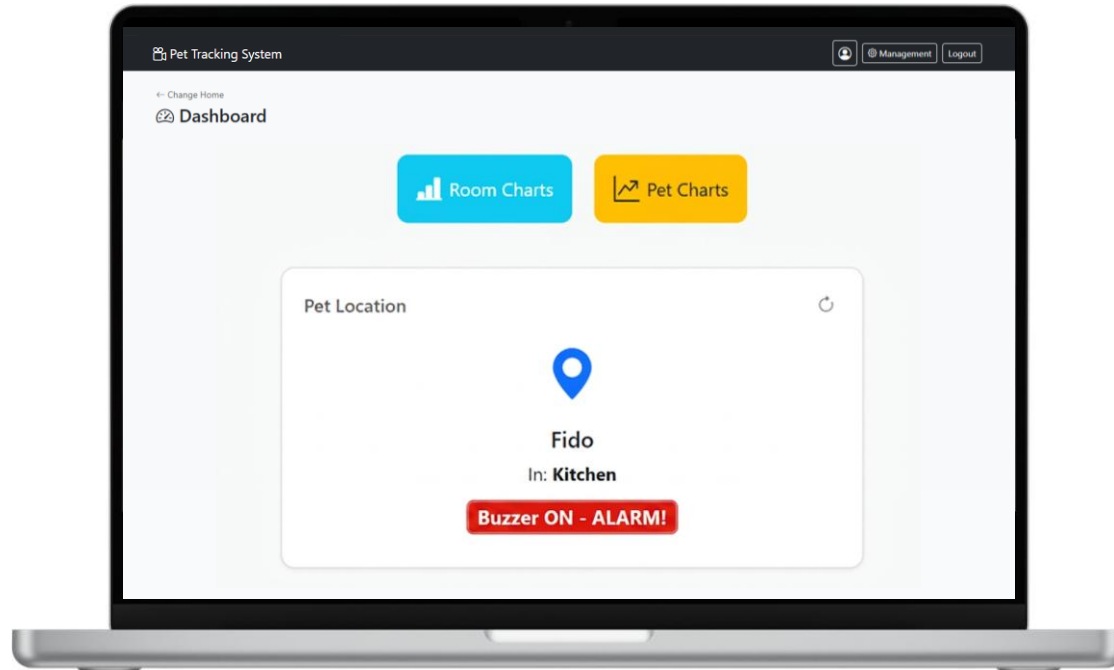
/buzzer



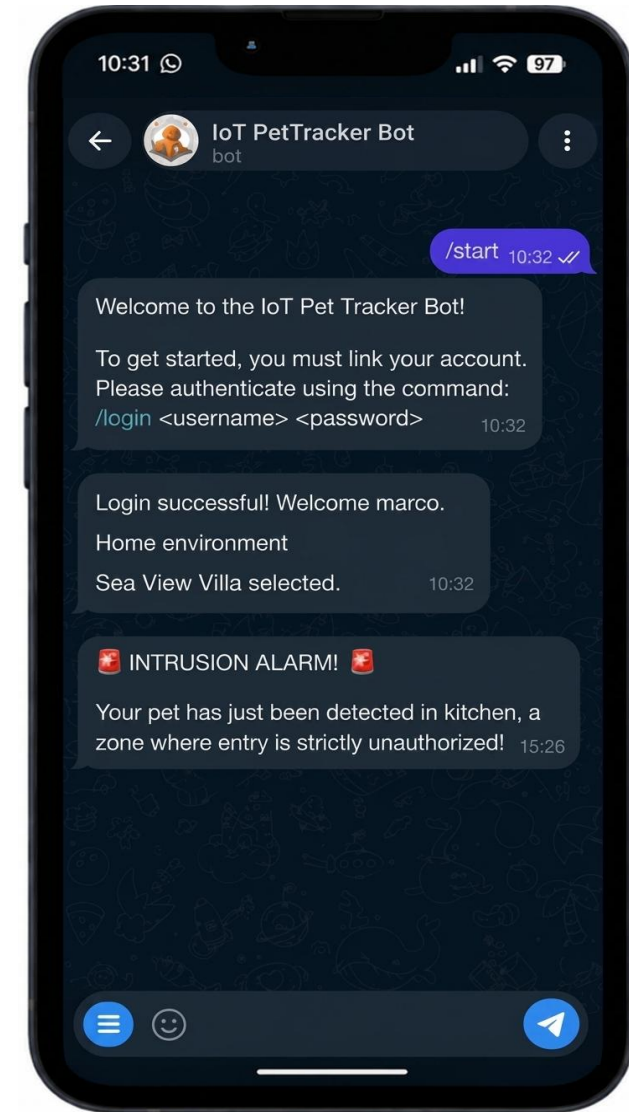
Only the administrator
can activate/deactivate the
buzzer when the pet is in
an allowed room



The Telegram Bot - Intrusion into a forbidden room



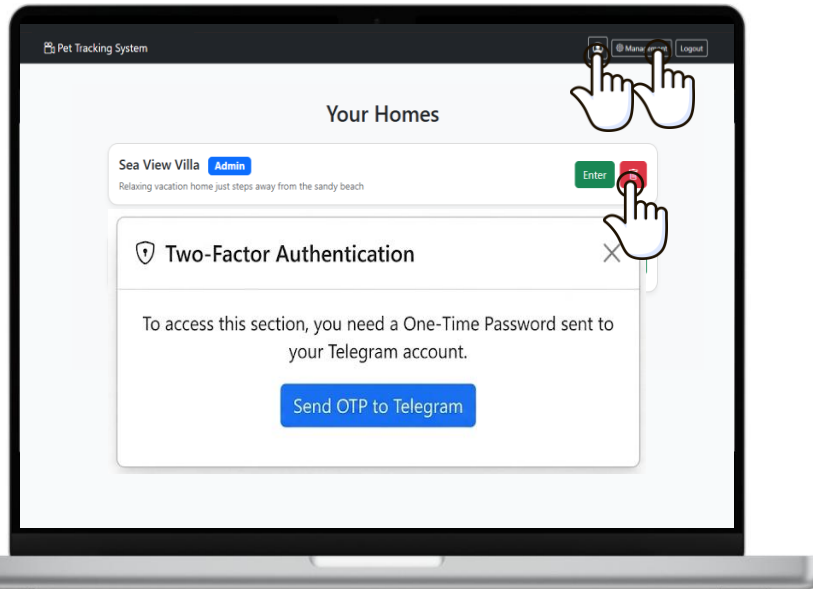
When the **pet enters a forbidden room**, the system automatically pushes a notification to all the logged users.



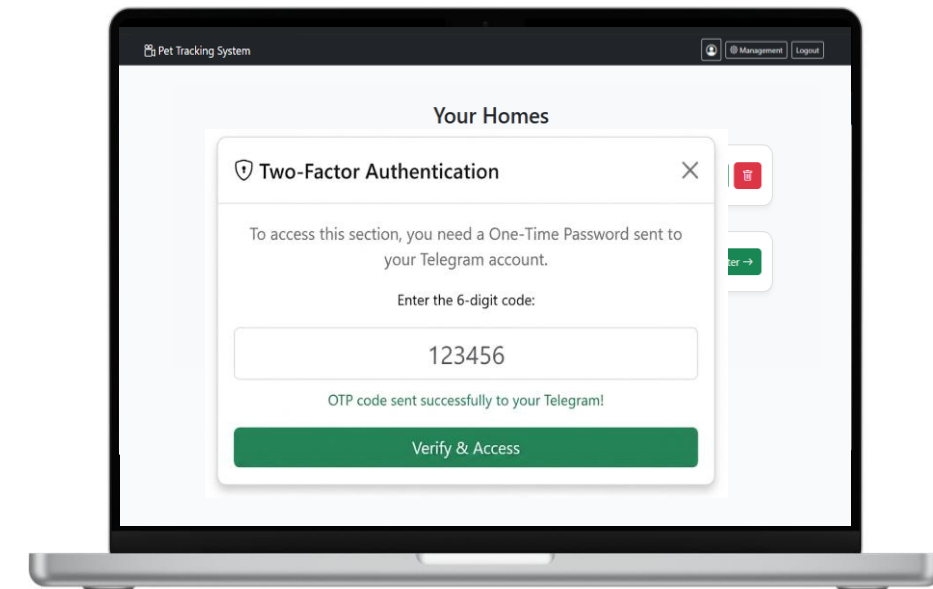
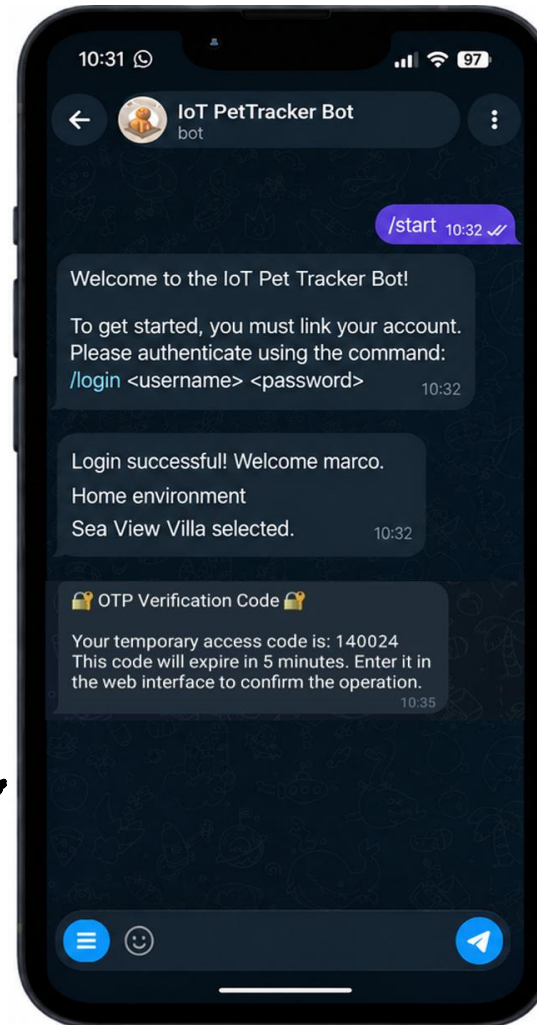
The Telegram Bot – Critical Operations

CRITICAL OPERATIONS

- Delete the Home Environment
- Access to the Management section
- Access to the User Profile



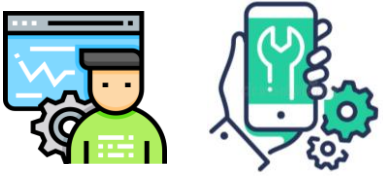
After pressing the "Send OTP to Telegram" button, the OTP verification code is sent to the bot



The 6-digit code has to be inserted here



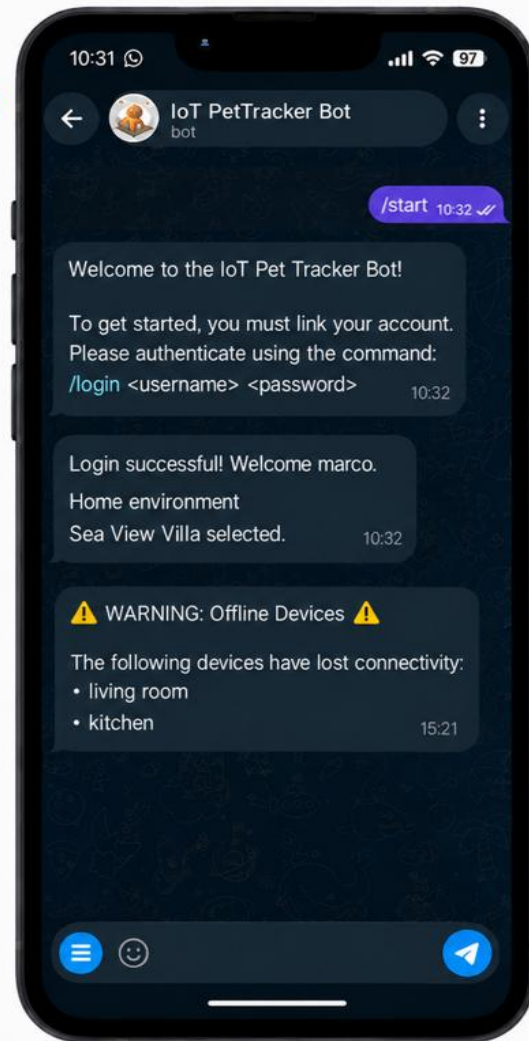
The Telegram Bot- State System Notifications



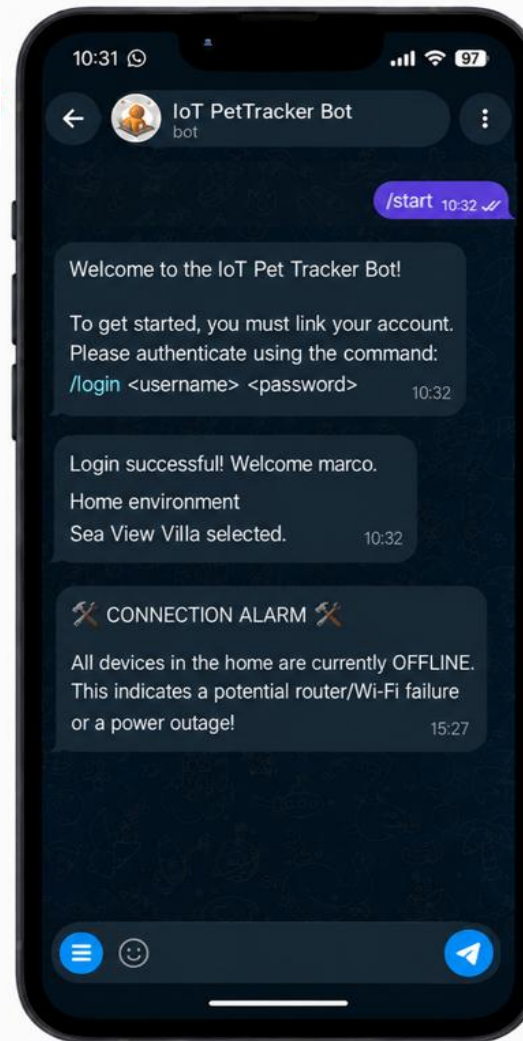
Users know the system's connection status in real time.



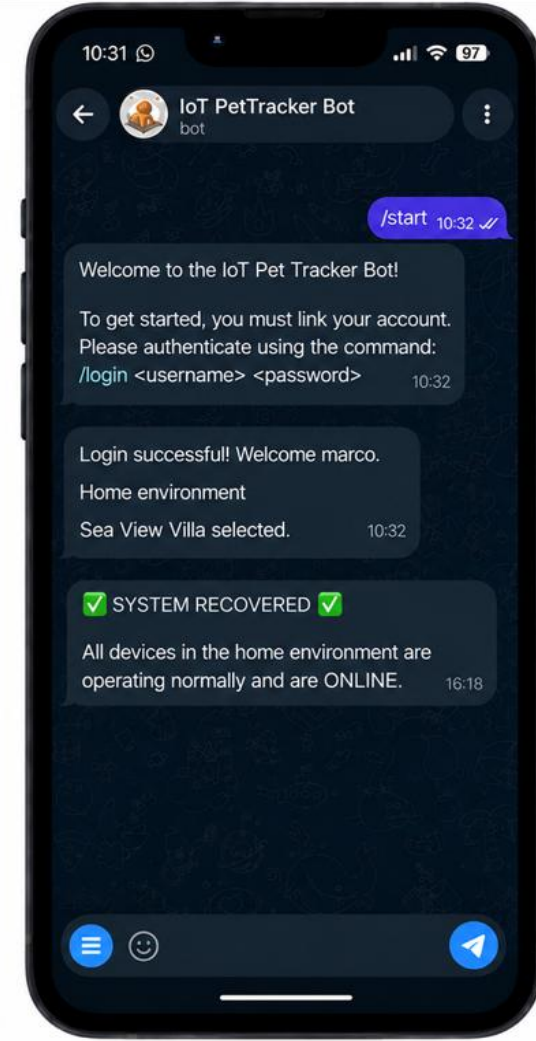
A



B



C



Pet-Tracker

Thanks for the attention!

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